

```
$ python3 day01.py
```

Python Day01

0000000000

```
>>> print("00000000000000")
```

```
00000000000000
```



```
$ cat agenda.md
```



1. Python Day01 — □□□□□□□□□□

2. □□□□

3. □□

4. □□□□

5. □□□□□

6. □□□□□□□□

7. Output

8. □□□□□□□

9. f-string □□□□□

10. Input

11. □□□□

12. □□□□

13. □□□□□□□□□□

transition: slide-up



Python

(type)

10

int
□□

12.3

float
□□□

'hi'

str
□□

True

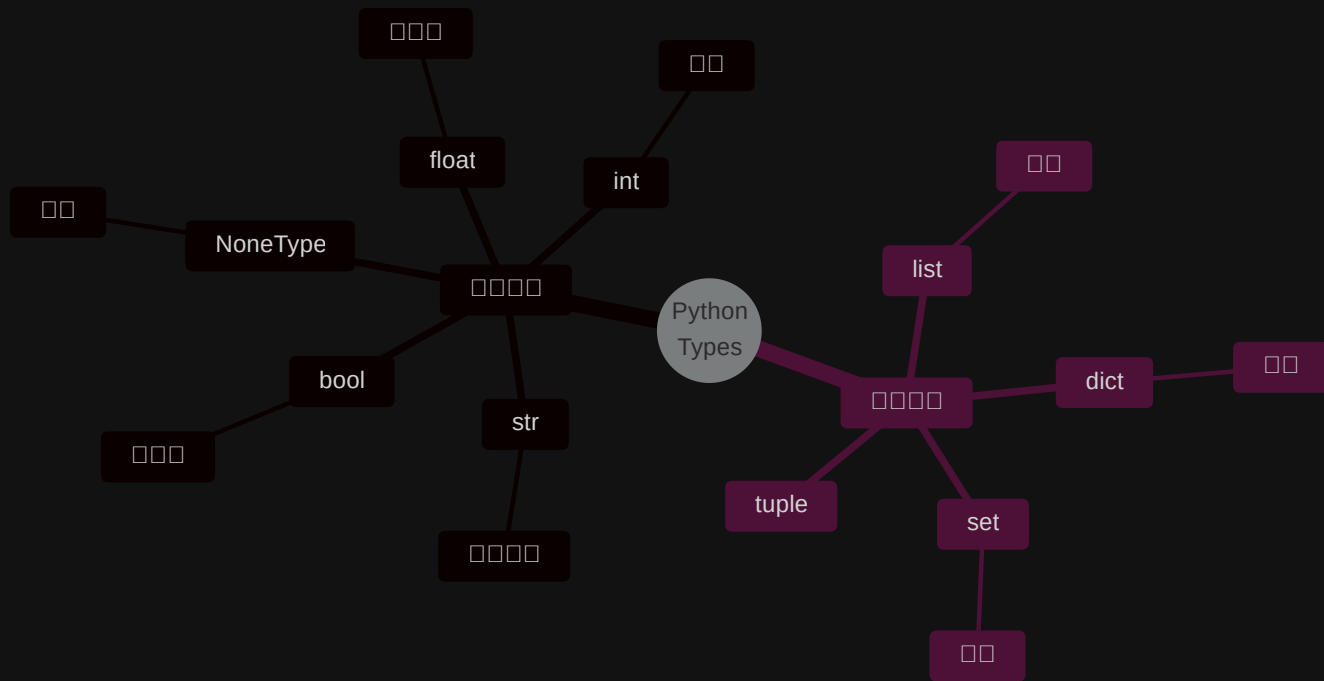
bool
□□

None

NoneType
□□

```
>>> type(10) # <class 'int'>
>>> type('hi') # <class 'str'>
```

transition: slide-up



transition: slide-up



000000

'123' ~ 00

123 ~ 00

'123' == 123 → False

'123'



123



>>> '123' == 123

False # 0000000000








































🤔

1. $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

2. $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

3. $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

4. $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

5. $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

6. $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

7. $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

8. $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

9. $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

10. $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

11. $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

12. $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

13. $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

14. $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

15. $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

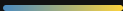
16. $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

17. $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

18. $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

19. $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

20. $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

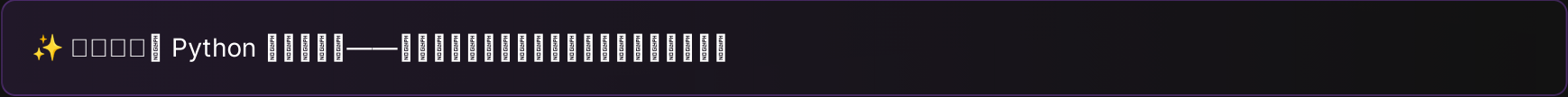


□ □ □ □ □ □ □ □ □ □ □ □



□ □ □ □ □ □ □ □ □ □

if, for, while



PART 03

Output

```
❏ print() [] [] [] [] []
```

transition: slide-up



```
1 print('hi')
2 print('hi')      # 00000000 hi 000
```

00000

```
1 hi
2 hi
```



```
1 # 🤔 000 - 000
2 name = 'Lucas'
3 age = 18
4 print('Hello,', name, 'you are', age, 'years old')
```

```
1 # 😊 0 f-string - 0000
2 name = 'Lucas'
3 age = 18
4 print(f'Hello, {name}! You are {age} years old.')
```

```
1 # 🤖 f-string 0000000
2 name = 'Lucas'
```



print()



□□□□ □□□□ □□□□ □□□□

```
1 print('hi')
2 print('hi')
3 # :
4 # hi
5 # hi
```

🔑 end — '\n' · sep — ' ' □

f-string □□□□□

f-string □□□□□□□□□□□□□□□□

```
1  # 0000
2  print(f'{{}}')
3
4  # 0000000000000000
5  pi = 3.14159265
6  print(f'π ≈ {pi:.2f}')    # π ≈ 3.14
7  print(f'π ≈ {pi:.4f}')    # π ≈ 3.1416
8
9  # 0000
10 print(f'{{42:05d}}')      # 00042
```

□□□□

```
{{:00}}
:.2f  - 0000
:05d  - 0000
```

 0000

□□□□□□□□□□□□

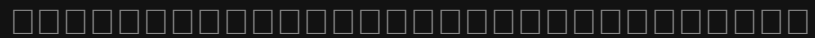
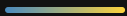
```
1  pi = 3.14159265
2  print(f'π ≈ {pi:.2f}')
3  print(f'π ≈ {pi:.4f}')
4  print(f'{{42:05d}}')
```

PART 04

Input

□□□□□□□□□□

transition: slide-up





`input()` □□□□□□□□□□□□

```
1 n = input()
2 nn = input('□□□□□□□')
```

⚠ `input()` □□□□□□□□□□□□□□□□
! '123' != 123



```
1 # □□□□□□□□□□
2 name = input('□□□□□ ')
3 print(f'Hello, {name}!')
4
5 # input() □□□□□□□□
6 age_str = input('□□□ ')
7 print(type(age_str)) # <class 'str'>
8
9 age = int(age_str) # □□□□
10 print(f'□□□□ {age + 1} □□')
```



□□□□□□□□□□□□□□□□

sys.stdin while + try □□□□

```
1  import sys
2
3  for line in sys.stdin:
4      line = line.strip()
5      print(line)
```

 sys.stdin

□□
□□□□□

 while+try

□□□□
□□□□□□

 □□□□

□□□
□□□□□□

transition: slide-up



 a.py 

```
1 a = input()
2 b = input()
3 print(a, b)
```

 in.txt 

```
1 Everything will get better
2 12345
```



```
1 python a.py < in.txt
```



```
1 Everything will get better 12345
```



□□□□□□□□

  in.txt □□□□□□□□□□□□□□□□

```
1 import sys
2
3 # UTF-8
4 sys.stdin.reconfigure(encoding='utf-8')
5
6 for line in sys.stdin:
7     line = line.strip()
8     print(line)
```

□□□□□□□□

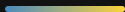
```
1 python a.py < in.txt > out.txt
```

 UTF-8 Windows□□□□□□

transition: slide-up



□□	□□□	□□	□□□□
□□	<code>n + 2</code>	Addition	<code>n=10 → 12</code>
□□	<code>n - 2</code>	Subtraction	<code>n=10 → 8</code>
□□	<code>n * 2</code>	Multiplication	<code>n=10 → 20</code>
□□	<code>n / 2</code>	Division integer division float	<code>n=10 → 5.0</code>
□□□□	<code>n // 2</code>	□□□□□□□□	<code>n=10 → 5</code>
□□	<code>n ** 6</code>	Power	<code>n=10 → 1000000</code>
□□□	<code>n % 6</code>	Modulo □□□□□□□□□□	<code>n=10 → 4</code>



```
1 n = 10
```

```
n + 2 → 12
```

```
n - 2 → 8
```

```
n * 2 → 20
```

```
n / 2 → 5.0 ← float
```

```
n // 2 → 5
```

```
n ** 6 → 1000000
```

```
n % 6 → 4 ← 10 ÷ 6 商
```

💡 / `float` // `float`

□ □ □ □ □ □ □

□ □ □ □ □ □ □ □ □ □

□ □ □ □ □ □ □ □ □ □ □

```
1  x = 10
2  x = x + 5    # x 15
3  x = x * 2    # x 30
4  print(x)
```

⚡ x += 1 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

□□□□□□□□

□□□□□□□□□□

■ ✓ True □□□

■ ✗ False □□□

□□□	□□
<	□□
<=	□□□□
>	□□
>=	□□□□
==	□□
!=	□□□

□□

```
1 10 <= 60      # True
2 123 == 123     # True
3 20 == 40       # False
4 '123' == 123   # False ⚠
```

⚠ '123' 123
False

Logical Operators

Logical Operators

Operator	Description
<code>and</code>	Both conditions must be true
<code>or</code>	At least one condition must be true
<code>not</code>	Reverse the result of the condition

```
1 age = 18
2 has_id = True
3
4 age >= 18 and has_id    # True
5 age < 18 or has_id     # True
6 not has_id              # False
```

🧠 `and` = Both conditions must be true · `or` = At least one condition must be true · `not` = Reverse the result of the condition

Truthy / Falsy □□□□□□

Python True / False □□□□□□

```
1 bool(0)           # False
2 bool(1)           # True
3 bool("")          # False
4 bool("hi")        # True
5 bool(None)        # False
6 bool([])          # False
```

Falsy False

- 0
- " "
- None
- []
- {}

Truthy True

- 0
-
-

if □□□□

□□□□□□□□

```
1 age = 18
2
3 if age >= 18:
4     print("□□□□")
```

⚠ Python □□□□□□□□□□□□□□□□

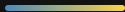
if / elif / else

□□□□□□

```
1  score = 85
2
3  if score >= 90:
4      print("A")
5  elif score >= 80:
6      print("B")
7  elif score >= 70:
8      print("C")
9  else:
10     print("F")
```

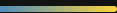


if score >= 90: print("A")
elif score >= 80: print("B")
elif score >= 70: print("C")
else: print("F")



```
1 username = input("000")
2 password = input("000")
3
4 if username == "admin" and password == "1234":
5     print("0000")
6 else:
7     print("0000")
```

🔑 000000input + 00 + and 0000



```
1 n = int(input("00000000"))
2
3 if n % 2 == 0:
4     print("00")
5 else:
6     print("00")
```

[illegible]



- Python □□□□□int / float / str / bool / None

- `input()` □□□□□□□□

- `print()` □□□□□ f-string □□□

- `/` □ `//` □ `%` □□□□□

- □□□□□□□□ + □□□□□□□□

- `if` / `elif` / `else` □□□□

